

Problem Set 3

E. Points

tianzheyu submissions							
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#	When	Who	Problem	Lang	Verdict	Time	Memory
379309992	Jun/18/2026 16:49UTC+10	tianzheyu	D - Points	C++23 (GCC 14-64, msys2)	Accepted	1062 ms	75800 KB

Process

The question is saying that we have a Cartesian coordinate system where (0,0) is located in the bottom-left corner. We have 3 operations:

1. add: we add point (x, y) into our coordinate system
2. remove: remove point (x, y) from out coordinate system
3. find: we return the point strictly above and strictly to the right of point, we will pick the leftmost one, if it is not unique, we will chooses the bottommost one.

Challenges and Overcoming

When coding my segment tree, I ran into a bug where my search function wasn't correctly pruning the search space for points strictly to the right of the query. I found the issue when I traced my base cases and realized my boundary condition, `tree[root].range.second <= x`, was misaligned with the 1-based index I generated during coordinate compression. Next time, to prevent this kind of off-by-one error with strict inequalities, I will explicitly write down the logical bounds (e.g. $X_{node} \leq X_{query}$) on paper before translating them into recursive return conditions.